

General information

Title : LED driver

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Master description revision : v2 (09/14/2025)

Change sheet

When?	What?
12/02/2025	Created the document, filled in basic info. Concept description and initial analysis.
12/02/2025	Define HLR for HW & FW, define LLR for FW, define block schematic.

References

Master description file

https://github.com/DevxMike/kitchen_driver/blob/main/Master_description_file_v2.pdf

General information	1
Change sheet	1
References	1
Brief	3
Requirements towards hardware	3
Requirements towards firmware	3
Detailed hardware description	4
Low-level requirements	5
Block schematic of electronic board	6
Detailed firmware description	7
Communication protocol	7
Packet structure	7
Low-level requirements	7

Brief

This document describes the flexible low-level driver part of the project. As per the frozen config (see the references) this block should be implemented as an independent part, independent embedded system with basic functionality and communication port. This section describes high-level requirements towards hardware and firmware sourcing from system requirements in master config file.

Requirements towards hardware

HLR_ID	SYS_ID	Description
HLR_HW_1	SYS_2 SYS_5	Board must utilize UART or USB port for communication purposes.
HLR_HW_2	SYS_3	As per details in the master config, the board must handle power regulation and distribution.
HLR_HW_3		Board must be able to power an SBC, LED and logic.
HLR_HW_4	SYS_4	Board must be able to exchange data with LED strip (WS2811 driver).
HLR_HW_5		Board must introduce elements enabling user to interact with.

NOTE: Because this project is a separate independent embedded system, it is assumed, that **SYS_1** requirement is satisfied.

Requirements towards firmware

HLR_ID	SYS_ID	Description
HLR_FW_1	SYS_2 SYS_5	Firmware must use the same packet description as the server side. It must come with methods for serializing and deserializing packets to act accordingly.
HLR_FW_2		Firmware must handle handshake and link status periodically.
HLR_FW_3		Firmware must handle packets containing settings from the server device.
HLR_FW_4	SYS_4	Firmware must handle communication with LED strip (WS2811 driver)
HLR_FW_5		Firmware must handle basic animations and functionalities (ON, OFF, BRIGHTNESS / COLOUR SETTING).
HLR_FW_6		Firmware must handle user interface.

Detailed hardware description

As per the master description, the board should incorporate STM32 microcontroller. For that, STM32L152 series microcontroller will be used as it introduces more than enough:

- 80kB RAM
- 11 timers
- USB and USART ports.

This part comes in both standalone chip and on nucleo development board hence makes the solution easy to develop, debug and port prototype to a board.

Low-level requirements

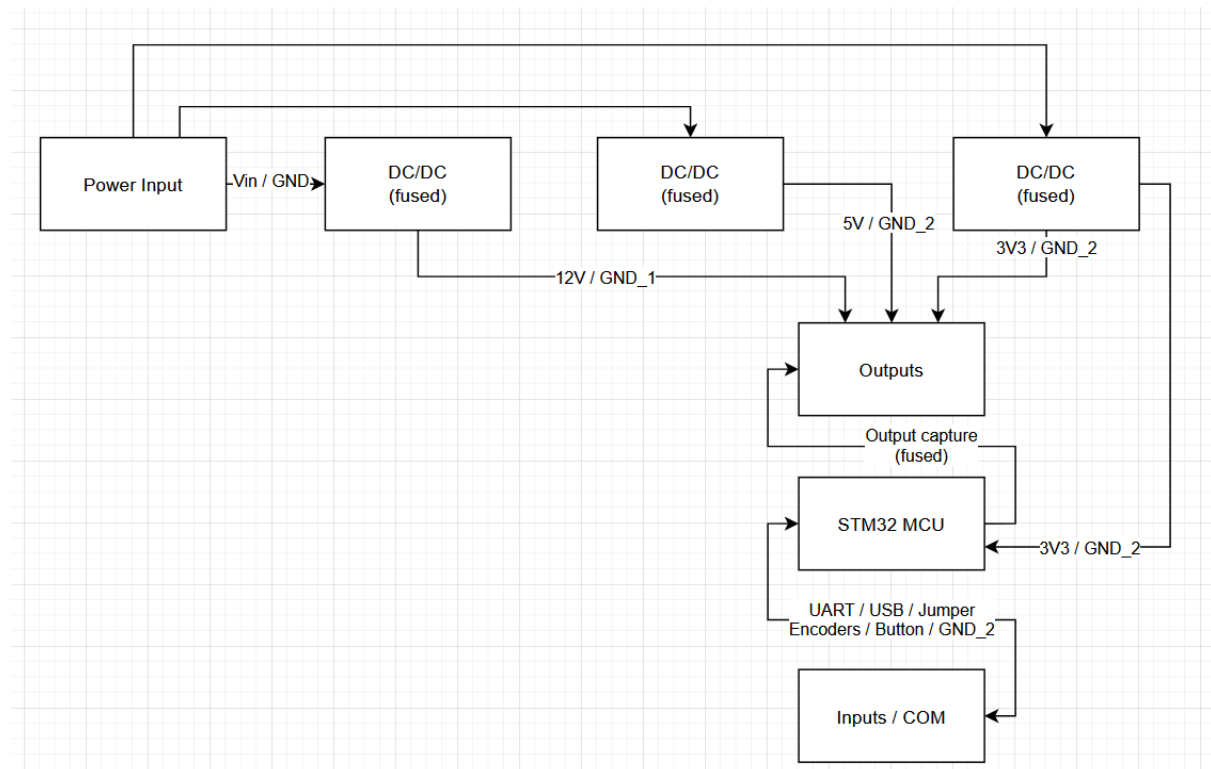
LLR_ID	HLR_ID	Description
LLR_HW_1	HLR_HW1	Board must introduce a jumper to select active port (USB; PA12 and PA11 or UART2; PA2 and PA3) by switching PB11 between GND and Vcc.
LLR_HW_2		If PB11 pin is shorted to GND, UART is enabled as active port.
LLR_HW_3		If PB11 pin is shorted to Vcc, USB is enabled as active port.
LLR_HW_4		UART should be configured as follows: 115200 baud, 8 data bits, 1 stop bit, parity none.
LLR_HW_5	HLR_HW_2 HLR_HW_3	Board must introduce impulse DC/DC converter to convert input voltage to 12V.
LLR_HW_6		12V line should be available in the output bar (supply for LED strip and auxiliary supply line).
LLR_HW_7		12V line should be minimum 2cm wide.
LLR_HW_8		Board must introduce impulse DC/DC converter to convert input voltage to 5V.
LLR_HW_9		5V line should be available in the output bar.
LLR_HW_10		5V line should be minimum 1cm wide.
LLR_HW_11		Board must introduce impulse DC/DC converter to convert input voltage to 3V3.
LLR_HW_12		3V3 line should be available in the output bar.
LLR_HW_13		3V3 line should be minimum 0.5cm wide.
LLR_HW_14		12V LED supply line should be fused up to 16A.
LLR_HW_15		12V auxiliary supply line should be fused up to 5A
LLR_HW_16		5V auxiliary supply line should be fused up to 5A.
LLR_HW_17		3V3 logic supply line should be fused up to 0.1A.
LLR_HW_18		3V3 auxiliary supply line should be fused up to 1A.
LLR_HW_19		Board must introduce a switch that allows physical power cutoff.
LLR_HW_20		3V3 logic supply (3V3 and GND) must be separated 12V supply lines.

LLR_ID	HLR_ID	Description
LLR_HW_20	HLR_HW_4	Output compare signal should be available in the output sections (TIMER 4, CHANNEL1, PB6).
LLR_HW_21		MCU should drive LEDs via optocoupler to ensure that supplies are separated.
LLR_HW_22	HLR_HW_5	Incremental encoder A should be attached to EXTI Line (PA8 PA9 encoder; PA10 encoder button).
LLR_HW_23		Incremental encoder B should be attached to EXTI Line (PB13 PB14 encoder; PB15 encoder button).
LLR_HW_24		PB11 input should be configured as a pull-up and available as a jumper.

Testing methodology

- electrical tests - shorts, supply availability
- functional tests - output compare oscilloscope test, encoder test, jumper test, USB and UART tests

Block schematic of electronic board



Detailed firmware description

Low-level requirements

NOTE: It has been decided, that communication between threads and task scheduling will be handled by freeRTOS framework.

LLR_ID	HLR_ID	Description